

Giorgi Merabishvili

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<https://giorgix121.github.io>

About Me

I am a first-year PhD student at North Carolina State University (NC State) advised by Professor Marcelo d'Amorim. I am broadly interested in software testing and improving robustness of software systems. Prior to my doctoral studies, I earned my master's degree in computer engineering from New York University. During my master's studies, I had the opportunity to work with Professor Andrea Stocco at the Technical University of Munich. I was funded by a DAAD research stipend.

Education

North Carolina State University , NC, US	2025 – Present
Doctor of Philosophy, Computer Science	GPA: N/A
Coursework: Design and Analysis of Algorithms Software Testing and Reliability	
New York University , NY, US	2023 – 2025
Master of Science, Computer Engineering	GPA: 3.80
Coursework: Machine Learning ML for Cybersecurity Deep Learning Probability and Stochastic Processes	
St. Francis College , NY, US	2019 – 2023
Bachelor of Science, Information Technology	GPA: 3.95
Coursework: Systems Analysis and Design Scripting Languages Networking Project Management	
Honors Thesis: "Robotics and Environmental Issues", Northeast Regional Honors Conference 2023	

Work Experience

Max Planck Institute for Security and Privacy	Bochum, Germany
Research Intern	Summer 2026
- Advisor: Dr. Marcel Böhme - Project: Automatically assess the correctness of LLM-generated security patches in the absence of a reliable test oracle.	
North Carolina State University	Raleigh, NC
Research Assistant	2025 – Present
- Advisor: Dr. Marcelo d'Amorim - Currently working in the area of game testing. I developed MR-guided VLM glitch detection technique for gameplay videos. - Worked on WebAssembly runtime analysis; transplanting regression tests across runtimes and using them as seed corpora for fuzzing.	
Technical University of Munich	Munich, Germany
Research Intern	2024 – 2025
- Advisor: Dr. Andrea Stocco - Conducted research on automated testing for deep learning systems, focusing on identifying boundary inputs. - Developed and implemented a method of latent space interpolation to minimize the distance from decision boundary. - Improved validity of generated pairs.	
RoboMaster NYU	New York, US
Computer Vision Engineer	2024 – 2025
- Worked on real-time object detection systems for tracking enemy robots. - Trained a model and tested real-time on robot provided by the mechanical team.	
Google Developer Student Club NYU	New York, US
Member, Technology Specialist	2023 – 2024
- Organized educational workshops with the GDSC team on machine learning and generative AI. - Managed technical resources and maintained documentation.	

Publications

- **Latent Regularization in Generative Test Input Generation**
G. Merabishvili, O. Weissl, A. Stocco.
In Proceedings of the 47th International Conference on Software Engineering Workshops, 2026
- **Targeted Deep Learning System Boundary Testing**
O. Weissl, A. Abdellatif, X. Chen, G. Merabishvili, V. Riccio, S. Kacianka, A. Stocco.
ACM Transactions on Software Engineering and Methodology, 2025 (TOSEM)
- **Left Behind, Not Forgotten: Reusing Regression Tests to Find Bugs in WebAssembly Runtimes**
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Under Submission, 2026

Honors and Awards

- **Recipient of DAAD Scholarship for the Research Project**
Munich, Germany, 2024
- **Merit Award Recipient at New York University**
New York, US, 2023-2025
- **Honors Program Scholar and Summa Cum Laude at St. Francis College**
New York, US, 2023
- **St. Francis College Travel Grant for Northeast Regional Honors Conference**
New York, US, 2023
- **Inducted to Sigma Beta Delta International Honor Society**
New York, US, 2023
- **Merit Award Scholarship and Institutional Scholarship Recipient at St. Francis College**
New York, US, 2019-2023

Activities

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| - Member of NCSU Water Polo Club, North Carolina, US | 2025 – Present |
| - Member of NYU Water Polo Club, New York, U.S | 2023 – 2025 |
| - NCAA Division I Water Polo for St. Francis College, New York, U.S | 2019 – 2020 |
| - Member of Water Polo Youth National Team and Club(Iveria), Tbilisi, Georgia | 2015 – 2019 |